Advanced Biotechnology

Lorenzo Di Michele
Professor of Bionanoscience
CEBT II - Michaelmas 2025



UNIVERSITY OF CAMBRIDGE

Lecture 12

Bioprocessing:
Primary isolation methods (part II) +
Product enrichment (liquid chromatography) +
(mention of) Final isolation methods



UNIVERSITY OF CAMBRIDGE

Lecture outline

1) Methods for primary isolation: Adsorption
2) Product enrichment (chromatography)
   - Principle and overview of chromatography
   - Types of liquid chromatography (brief description of principles)
   - Quantify differential migration
3) Mention of final isolation methods

Chapter 11 of Bioprocess Engineering Principles (11.9-11.12)

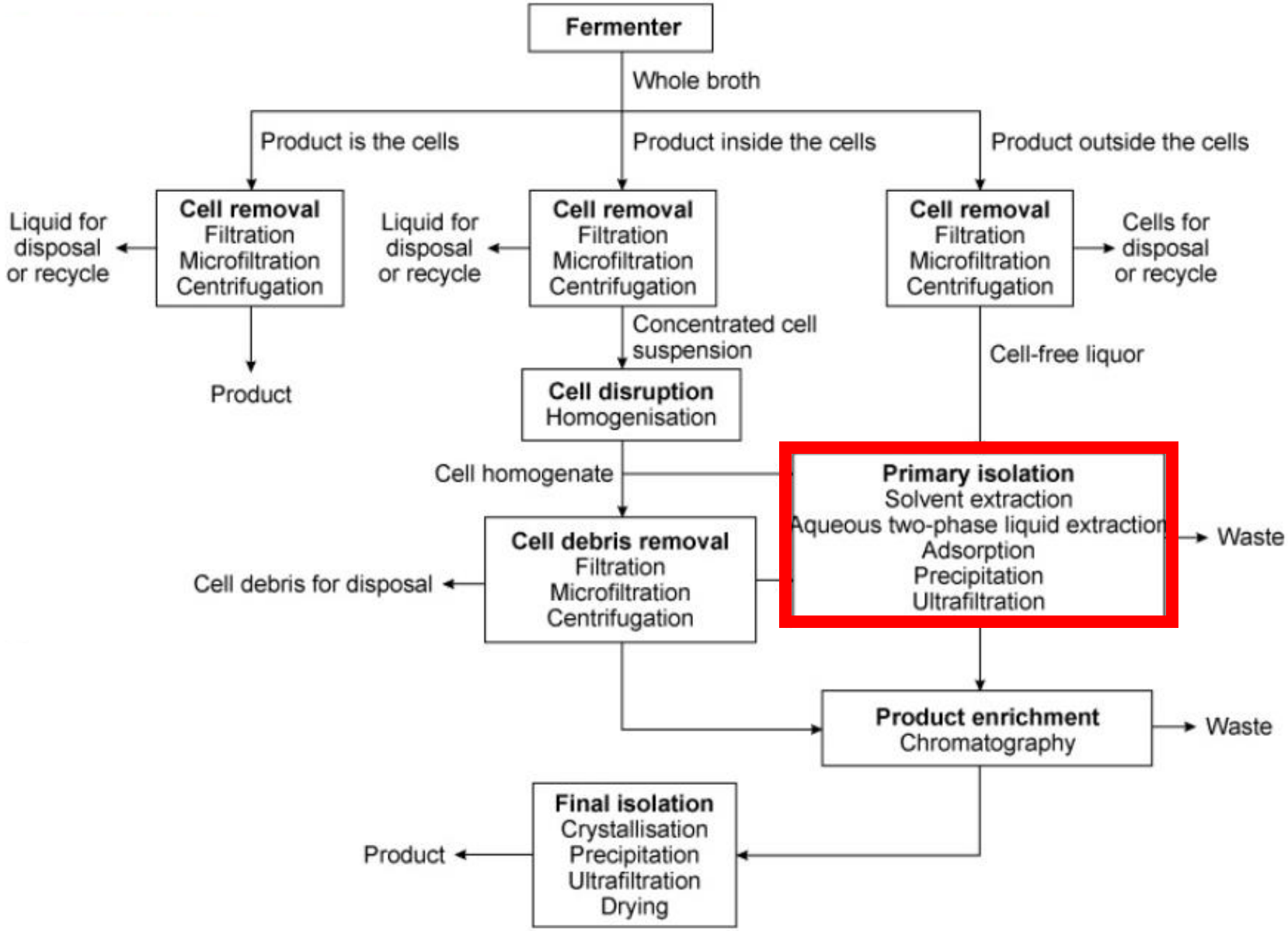
Learning outcomes

1) Describe principles and steps of isolation by adsorption
2) Describe and apply adsorption isotherms (Langmuir, Freundlich)
3) Describe principles of product enrichment with chromatography
4) Describe basic principles of liquid chromatography techniques (adsorption, partition, ion exchange, gel and affinity chromatography)
5) Describe relevant parameters (elution volume, partition coefficient, capacity factor, retention ratio)

Lecture outline

1) **Methods for primary isolation: Adsorption**
2) Product enrichment (chromatography)
   - Principle and overview of chromatography
   - Types of liquid chromatography (brief description of principles)
   - Quantify differential migration
3) Mention of final isolation methods

Downstream process logic



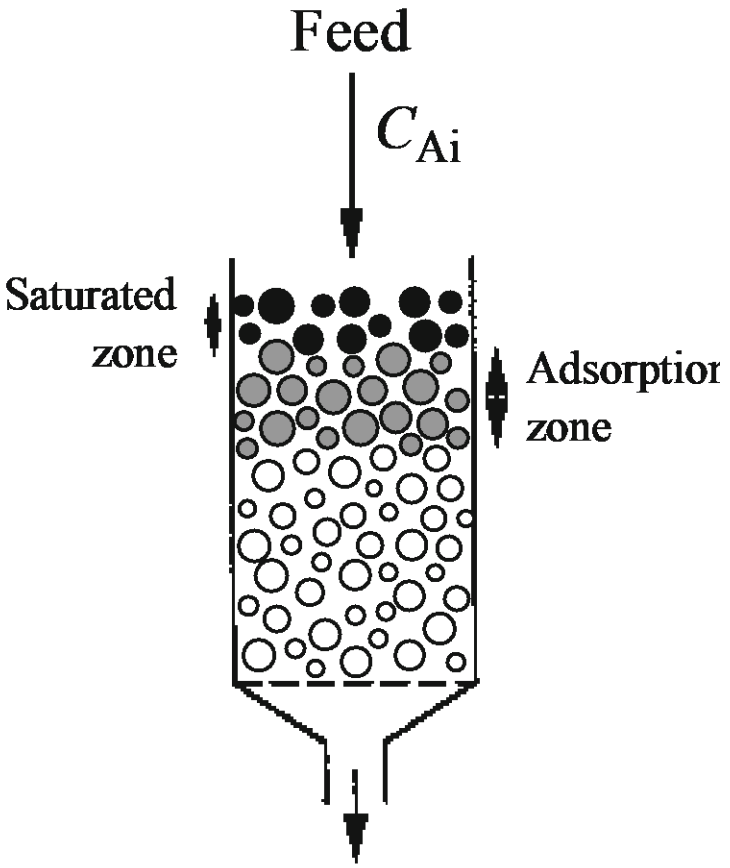
Primary Isolation methods

Typical operations include (but are not limited to)

I. Solvent extraction (last lecture)
II. Aqueous two-phase liquid extraction (last lecture)
III. Precipitation (last lecture)
IV. Adsorption (this lecture)

IV. Adsorption

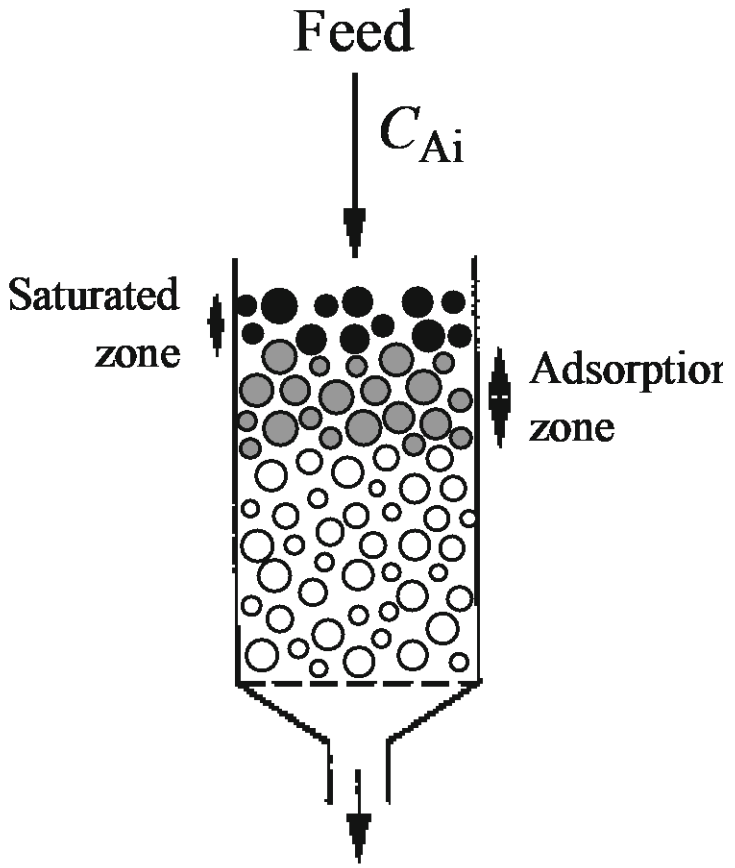
- Adsorption is a surface process where components of a gas or liquid bind to solid surfaces via physical or chemical interactions (van der Walls, electrostatic)
- The compound that binds is called **adsorbate**, while the material to which the adsorbate binds is called **adsorbant**
- Adsorption is often used as primary isolation method, particularly in pharma for the **separation of proteins, amino-acids, antibodies and enzymes**. It is also used in **the treatment of waste-waters**
- The adsorbant is normally **porous particles** with very high surface area for adsorption, packed in a column. Materials include **activated charcoal** and **resins** made from styrene, divinylbenzene, or acrylamide polymers



IV. Adsorption

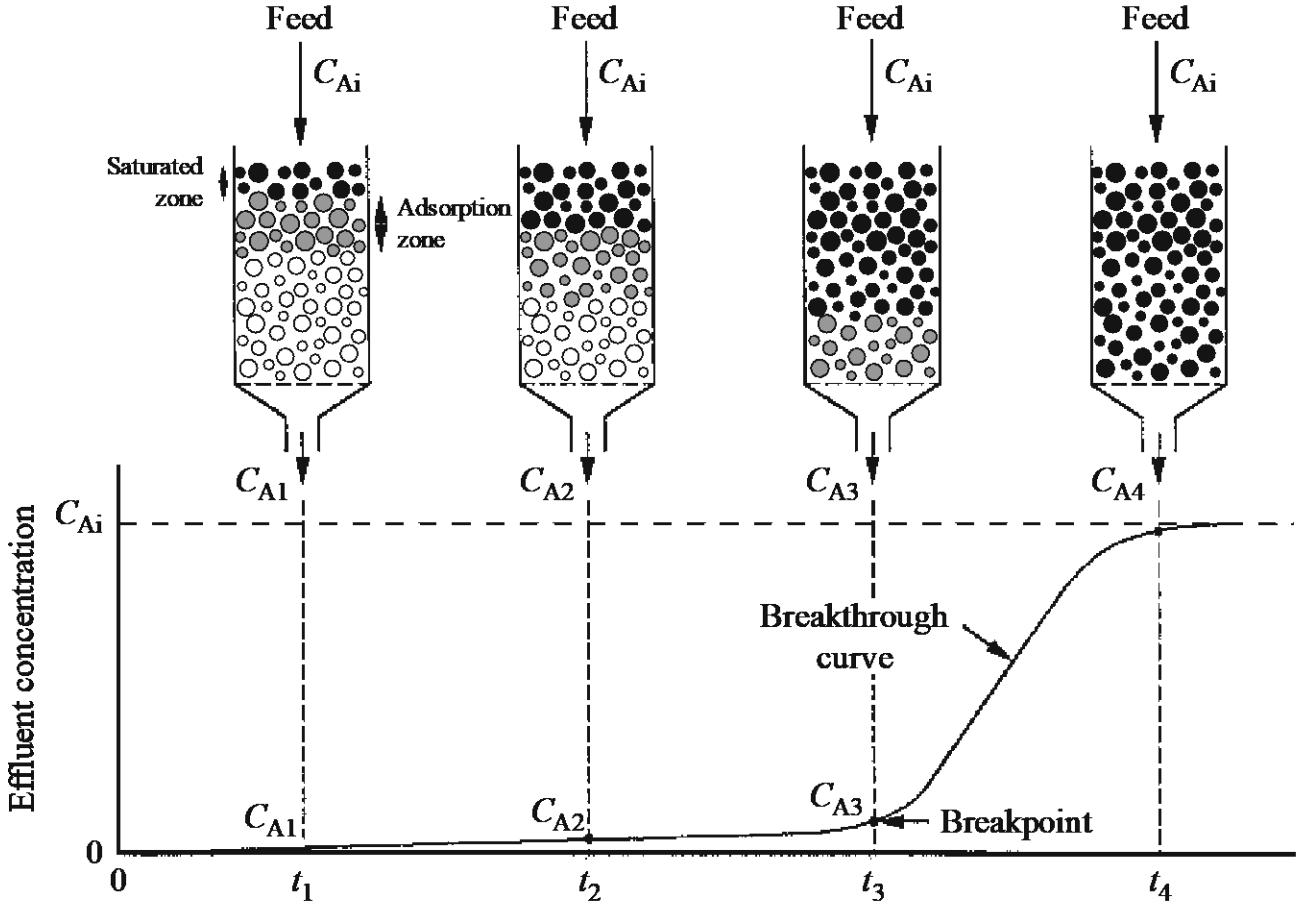
- There are four main stages in isolation by adsorption

1. **Binding**: the sample is contacted with the adsorbant, and the adsorbate binds
2. **Washing**: the adsorbant is washed to remove unbound and weakly bound compounds
3. **Elution**: the adsobate is made to unbind so it can be recovered. This is normally done by flowing a different solvent that reverses the interactions that lead to adsorption in the first place
4. **Regeneration**: the adsorbant is brought back to its original conditions so it can be re-used, for instance by replacing the elution solvent with the original binding solvent



Binding phase

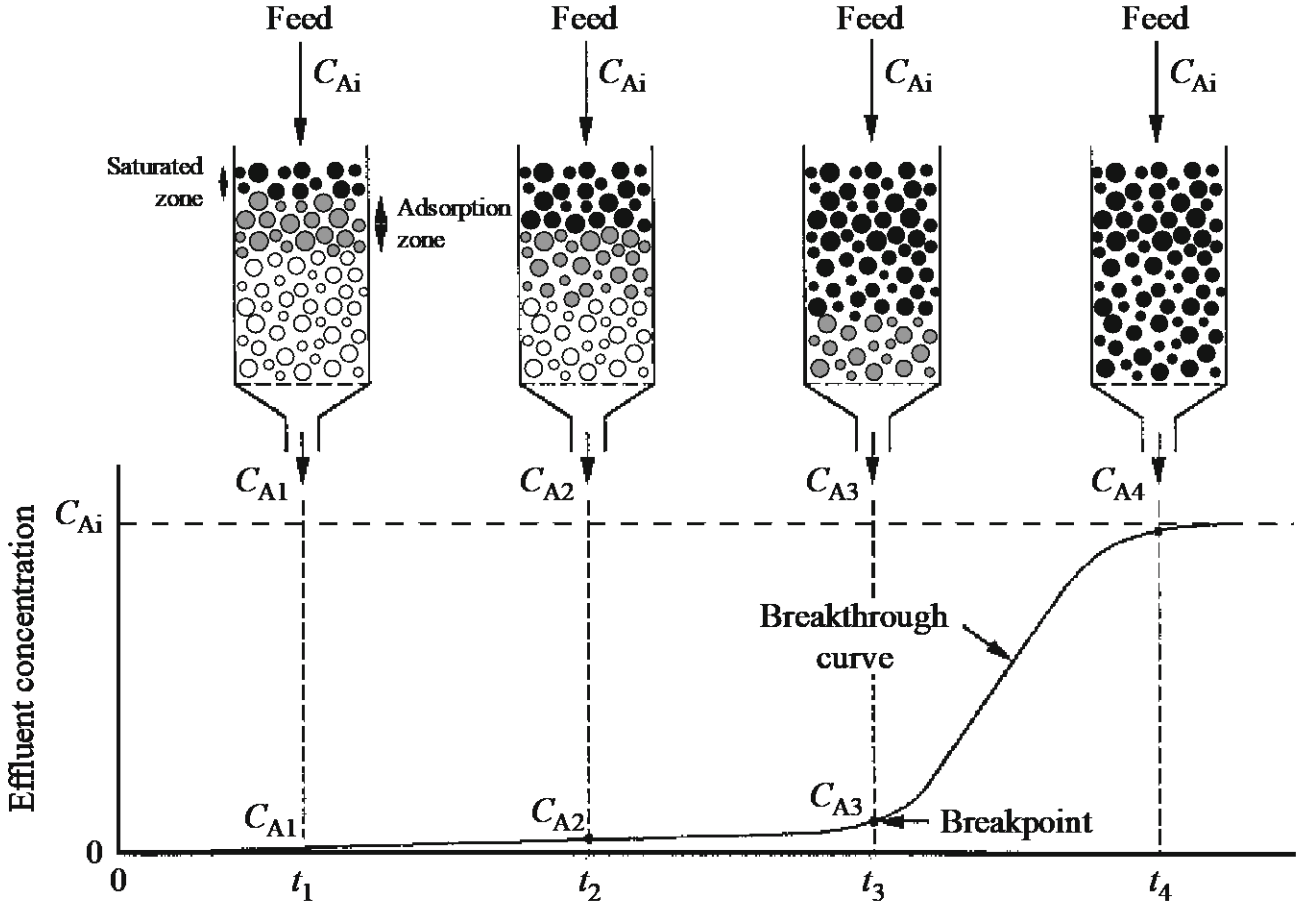
- An adsorption zone propagates downwards, at a rate slower than the liquid flow. In this zone, adsorption is occurring, and the resin is not yet saturated
- Above this zone, the resin is saturated
- Feed comes out from the other side of the column (effluent) with a time-dependent C_A



From P.M. Doran, "Bioprocess Engineering Principles" Academic Press, 2nd ed. 2012.

Binding phase

- At early times, when the adsorption zone has not yet reached the bottom of the colum, C_A is low
- At late times, the adsorption and saturation zones reach the bottom of the column. The column becomes saturated and C_A starts increasing
- The point when the C_A starts increasing is the "breakpoint"
- The process is usually terminated at this stage, to avoid wasting target compound in the effluent



From P.M. Doran, "Bioprocess Engineering Principles" Academic Press, 2nd ed. 2012.

Adsorption isotherms: Langmuir isotherm

- Describe adsorption processes in terms of the **fraction of adsorbate bound to the adsorbant** as a function of the **bulk concentration of adsorbate** at constant temperature
- The simplest form is the **Langmuir isotherm**, developed by Irving Langmuir (Nobel laureate for his contributions to surface chemistry in 1932)



Langmuir in 1932 from Wikipedia

Adsorption isotherms: Langmuir isotherm

$$C_{AS}^{*} = \frac{C_{ASm} K_A C_A^{*}}{1 + K_A C_A^{*}}$$

Where:

- C_{AS}^{*} is the equilibrium concentration of compound A (the absorbate) bound onto the adsorbant, expressed as mass (or moles) of absorbate over mass or volume of the (solid) absorbant
- C_{ASm} is the maximum attainable value of C_{AS}^{*} (the saturation level for the absorbant). $C_{AS}^{*}/C_{ASm} = \theta$ is thus the fraction of binding sites on the absorbant occupied by A
- C_A^{*} is the equilibrium concentration of A in the fluid phase, in units of $\mathrm{kg\,m^{-3}}$ or $\mathrm{mol\,m^{-3}}$
- K_A is a constant, in units of $\mathrm{kg^{-1}\,m^3}$ or $\mathrm{mol^{-1}\,m^3}$

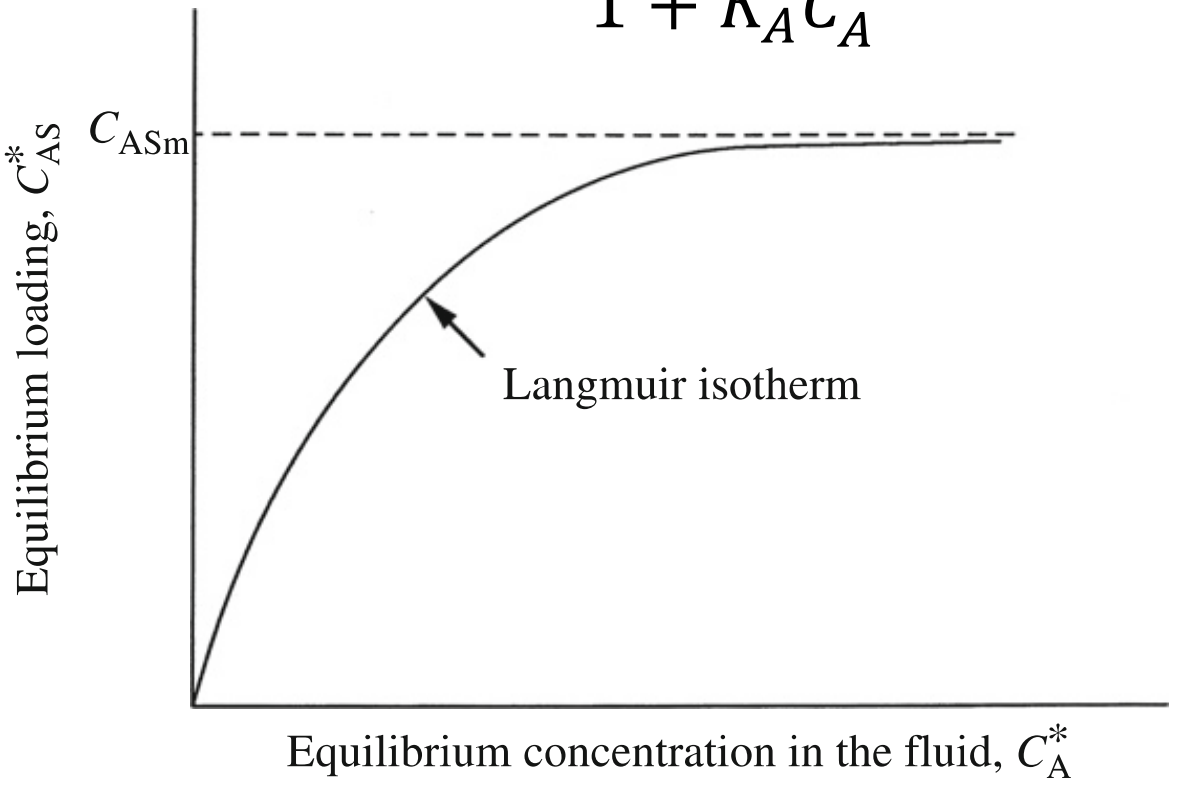
Adsorption isotherms: Langmuir isotherm

The Langmuir isotherm is **only valid** if the following **assumptions** are true:

- Adsorbed molecules only form a monolayer on the surface
- All binding sites on the surface are equivalent
- Interactions between adsorbed molecules are negligible

These **assumptions are often non verified in real-life scenarios**. For instance, in complex adsorption media binding sites are rarely all equivalent

$$C_{AS}^{*} = \frac{C_{ASm} K_A C_A^{*}}{1 + K_A C_A^{*}}$$

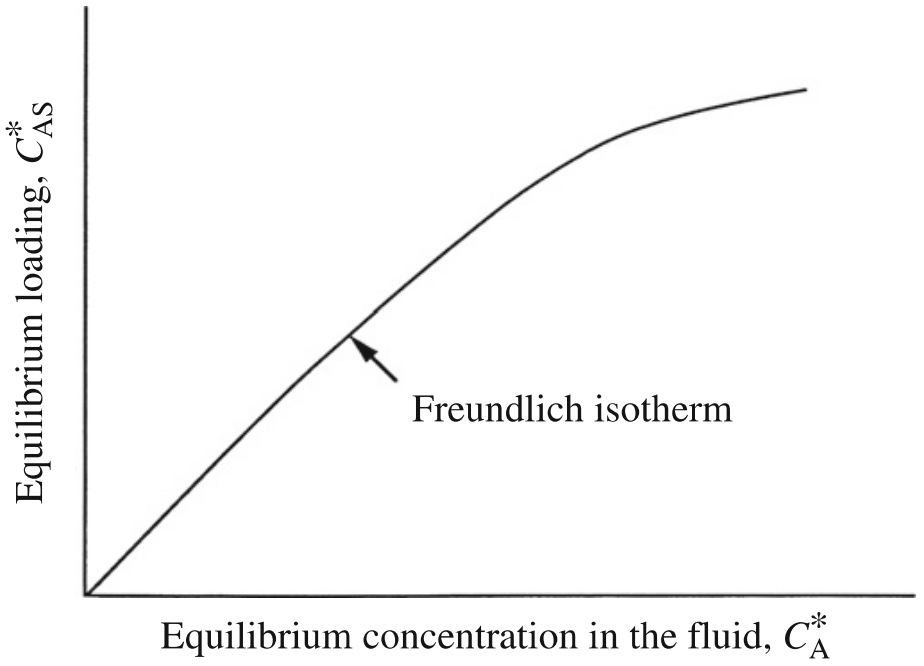


Langmuir isotherm from P.M. Doran, "Bioprocess Engineering Principles" Academic Press, 2nd ed. 2012.

Adsorption isotherms: Freundlich isotherm

- Other isotherms exist, applicable in different contexts under different assumptions
- Of particular interest in liquid/solid adsorption is the **Freundlich isotherm**, that works well for isolation of antibiotics, hormones and steroids
- n is a constant with $n > 1$
- K_f is a constant whose units depend on the units used for C_{AS}^* and C_A^* and the value of n
- Note that different from the Langmuir isotherm, the Freundlich isotherm does not reach saturation!

$$C_{AS}^* = K_f C_A^{*1/n}$$



Freundlich isotherm from P.M. Doran, "Bioprocess Engineering Principles" Academic Press, 2nd ed. 2012.

Example, antibody recovery by adsorption

Cell-free fermentation liquor contains $C_A = 8 \times 10^{-5}\ \mathrm{mol\,l^{-1}}$ immunoglobulin G. It is proposed to recover at least 90% of this antibody by adsorption on synthetic, nonpolar resin. Experimental equilibrium data are correlated as follows:

$$C_{AS}^{*} = 5.5 \times 10^{-5} C_{A}^{*0.35}$$

where C_{AS}^{*} is the moles of solute adsorbed per $\mathrm{cm^3}$ of adsorbant and C_A^* is the equilibrium liquid-phase solute concentration in $\mathrm{mol\,l^{-1}}$. **What volume of resin is required to treat V=2 $\mathrm{m^3}$ of fermentation liquor?**

Solution

The volume of resin needed is equal to

$$V_{res} = \frac{N_{AS}}{C_{AS}^*}$$

Where N_{AS} is the total number of moles of the antibody we aim to recover and C_{AS}^* is the number of moles adsorbed on the resin per unit volume of the resin.

To compute C_{AS}^*, we can use the correlation given. We start by computing C_A^*, or the concentration of antibody that remains in solution after adsorption. We know that we want to recover 90% of the antibody, so 10% will remain in solution.

$$C_A^* = \frac{1}{10} C_A = 8 \times 10^{-6} \ \mathrm{mol\ l^{-1}}$$

Solution

We can now compute

$$C_{AS}^{*} = 5.5 \times 10^{-5} \times (8 \times 10^{-6})^{0.35} = 9.05 \times 10^{-7} \mathrm{mol\,cm^{-3}}$$

Next, we calculate the total number of moles of adsorbed antibody

$$N_{AS} = 0.9 \times C_A \times V = 0.9 \times 8 \times 10^{-5}\ \mathrm{mol\,l^{-1}} \times 2\ \mathrm{m^3} \times \frac{1000\ \mathrm{l}}{\mathrm{m^3}}$$

$$N_{AS} = 0.144\ \mathrm{mol}$$

Solution

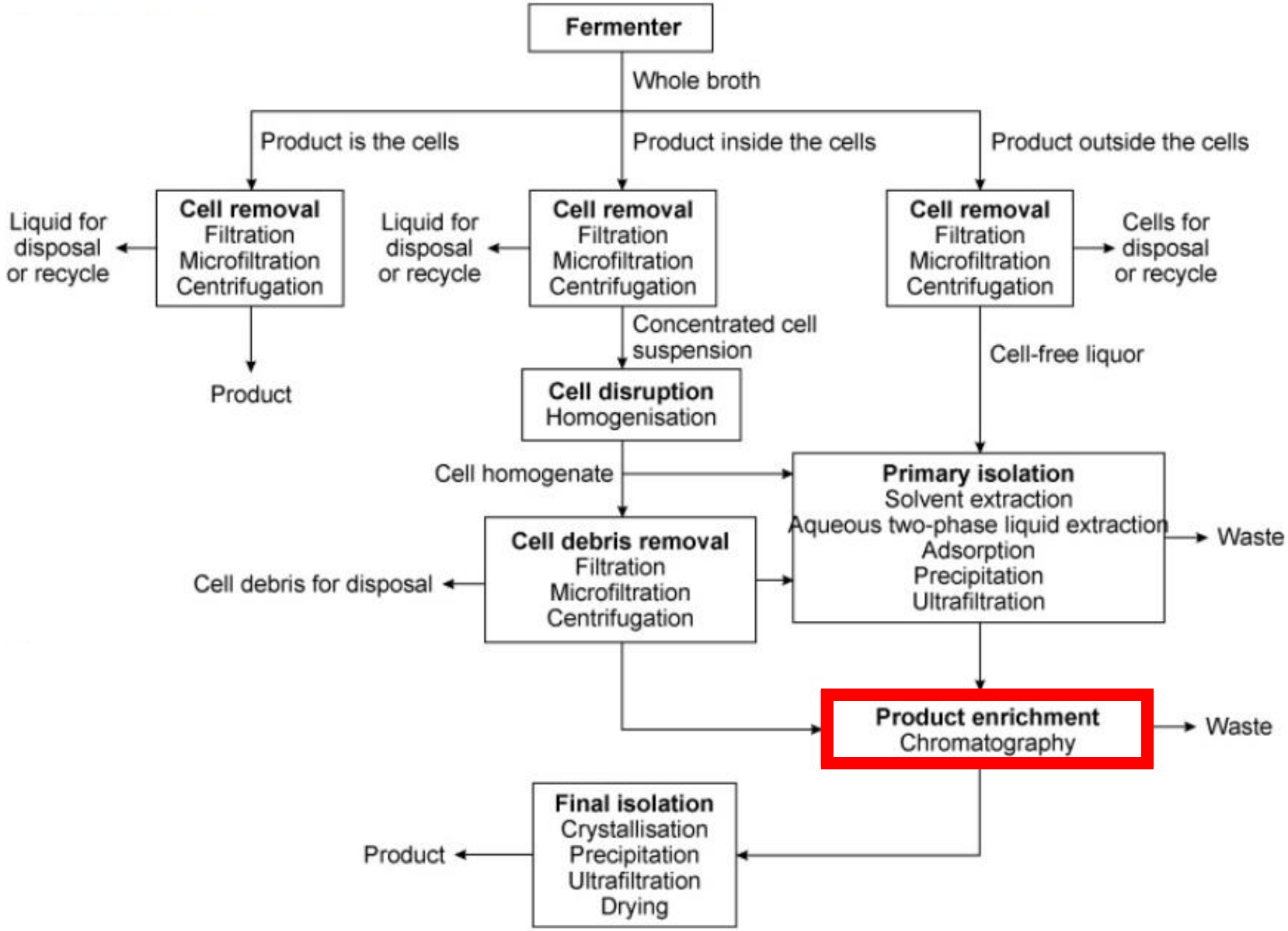
The volume of resin needed to capture this much antibody thus is thus

$$V_{res} = \frac{N_{AS}}{C_{AS}^{*}} = \frac{0.144\ \text{mol}}{9.05 \times 10^{-7}\text{mol cm}^{-3}} = 1.59 \times 10^{5}\text{cm}^{3} = 0.159\ \text{m}^{3}$$

Lecture outline

1) Methods for primary isolation: Adsorption
2) **Product enrichment (chromatography)**
   - **Principle and overview of chromatography**
   - **Types of liquid chromatography (brief description of principles)**
   - **Quantify differential migration**
3) Mention of final isolation methods

Downstream process logic

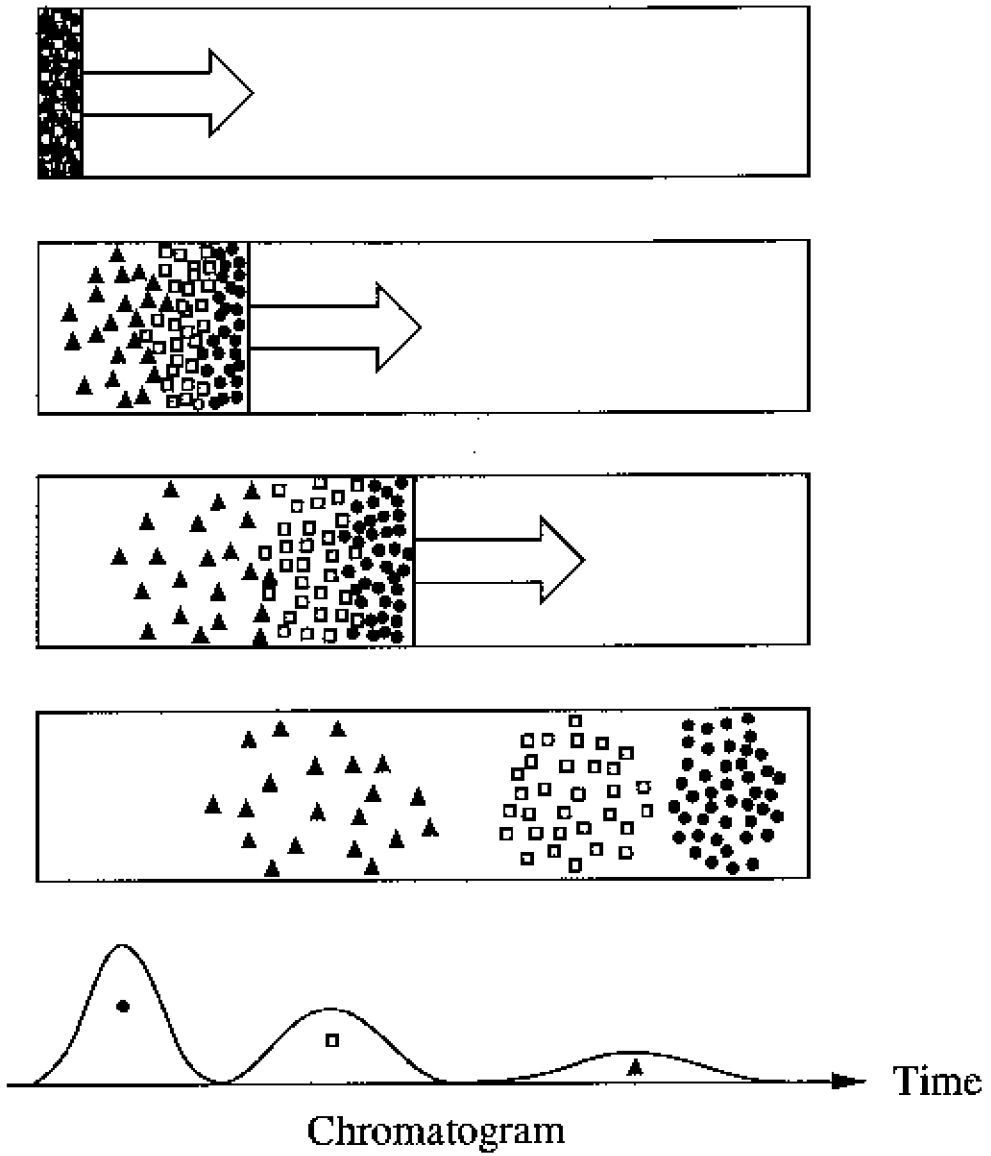


Product enrichment

- At this stage, we are left with a liquor with containing the target product which has been already **partially purified** and possibly concentrated at the primary isolation stage.
- However, because of the non-specific nature of many primary isolation processes, the **product has low purity** (*e.g.* other proteins besides the target protein may be present).
- Additionally, there may be compounds present in the liquor that have been introduced during the primary isolation.
- **Product enrichment further purifies the product by removing other compounds with similar physical and chemical properties which are difficult to remove with primary isolation**.
- Typically, product enhancement relies on **chromatographic techniques**

Liquid chromatography

- Resolves mixtures into individual components.
- Relies on **differential migration**:
  1. A sample is flown through a **column packed with resin using an eluent**
  2. Different components move through the resin at different rates
  3. They come out at different **elution times** on the other side, enabling separation
- The graph of concentration of the various components in the effluent vs elution time is the **chromatogram**
- The resin is the **stationary phase**
- The liquid that carries the solutes is the **mobile phase**

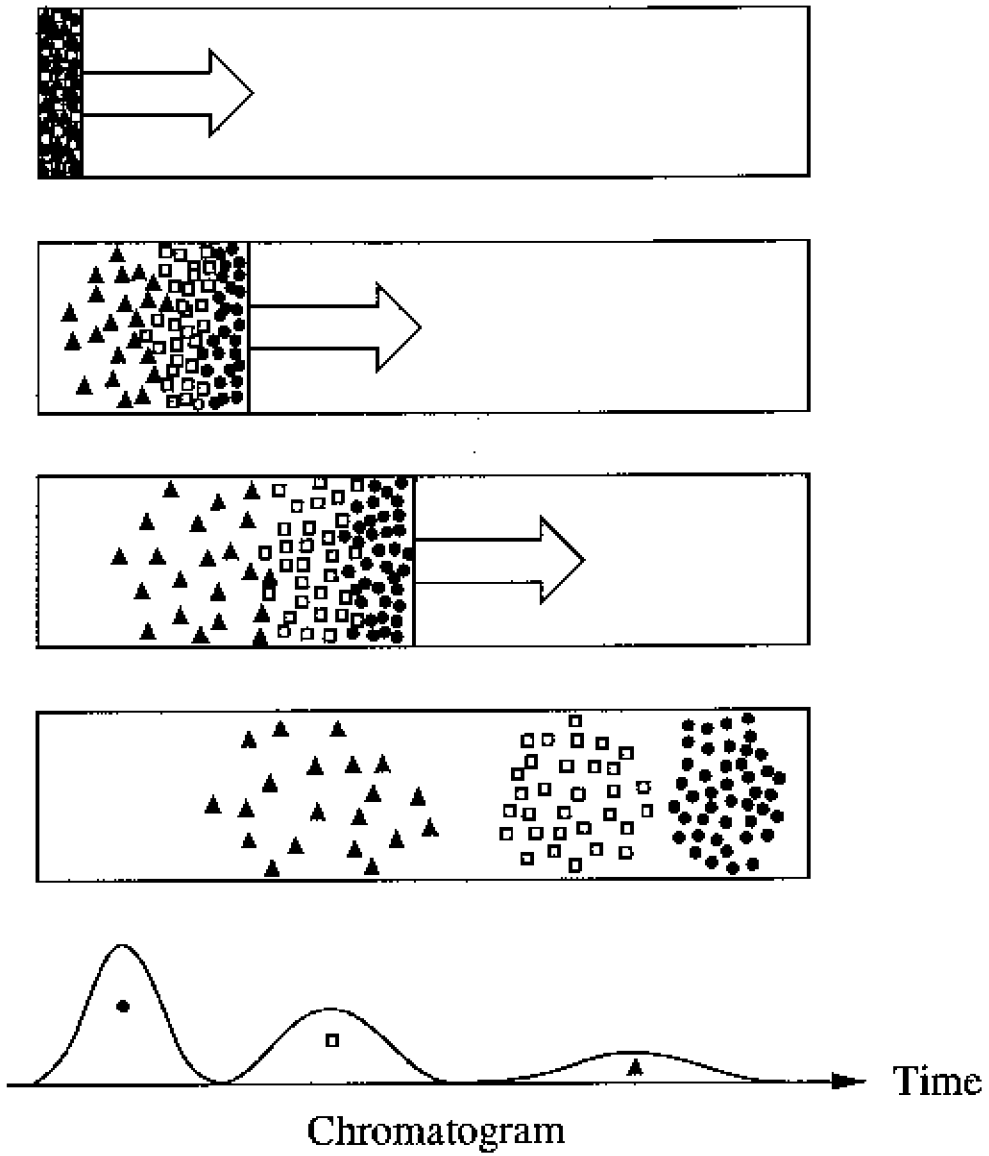


From P.M. Doran, "Bioprocess Engineering Principles" Academic Press, 2nd ed. 2012.

Liquid chromatography

- The mobile phase can be also gas, as in gas chromatography. However **liquid chromatography** is the one relevant to bioprocessing

- Liquid chromatography achieve high purity and is therefore commonly used for **enrichment of bioproducts in the pharmaceutical industry**, like **antibodies, recombinant proteins and nucleic acids** used as therapeutics



From P.M. Doran, "Bioprocess Engineering Principles" Academic Press, 2nd ed. 2012.

Types of liquid chromatography

There are five main types of liquid chromatography relevant to bioprocessing:

1) Adsorption chromatography
2) Partition chromatography
3) Gel chromatography
4) Ion-exchange chromatography
5) Affinity chromatography

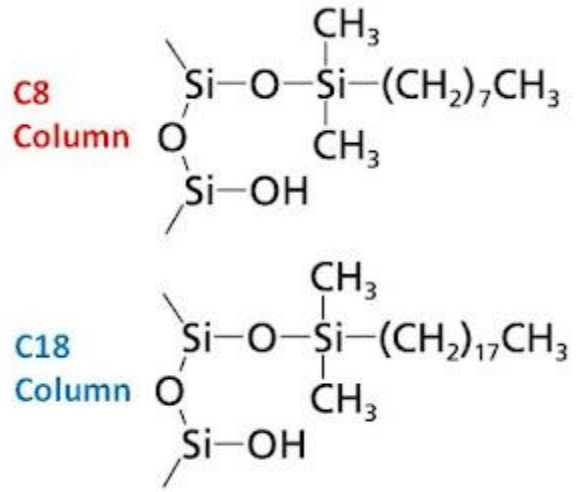
We will briefly review their principles

1) Adsorption chromatography

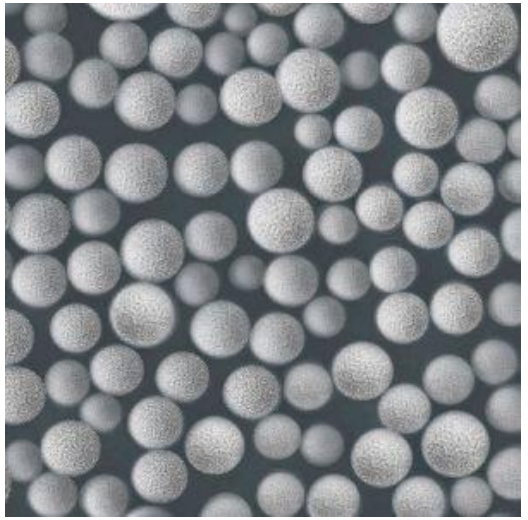
- Utilizes **non-specific adsorption** of compounds against the stationary phase to generate differential migration
- Stationary phase typically consists of particles of **silica gel**, **alumina**, **diatomaceous earth**, and **charcoal**
- Biological molecules adsorb on these materials typically driven by **electrostatic forces**
- Order of elution (order in which the compounds emerge from the column) depends primarily on **molecular polarity**
- **Eluent (mobile phase) polarity** is also important to determine degree of separation
- **Important: while principle is the same, adsorption chromatography is different from fixed bed adsorption used as a primary isolation method!**

2) Partition chromatography

- Principle leading to differential migration is differential partitioning of compounds between **two "liquid" phases**.
- One "liquid" phase is immobilized onto a support, such as silica, making it the stationary phase. A common stationary phase is **hydrocarbon with 8 or 18 carbons** bonded to silica gel: these materials are called **C8 and C18 packings, respectively**.
- The second liquid phase is the mobile phase. Common mobile phases are aqueous.



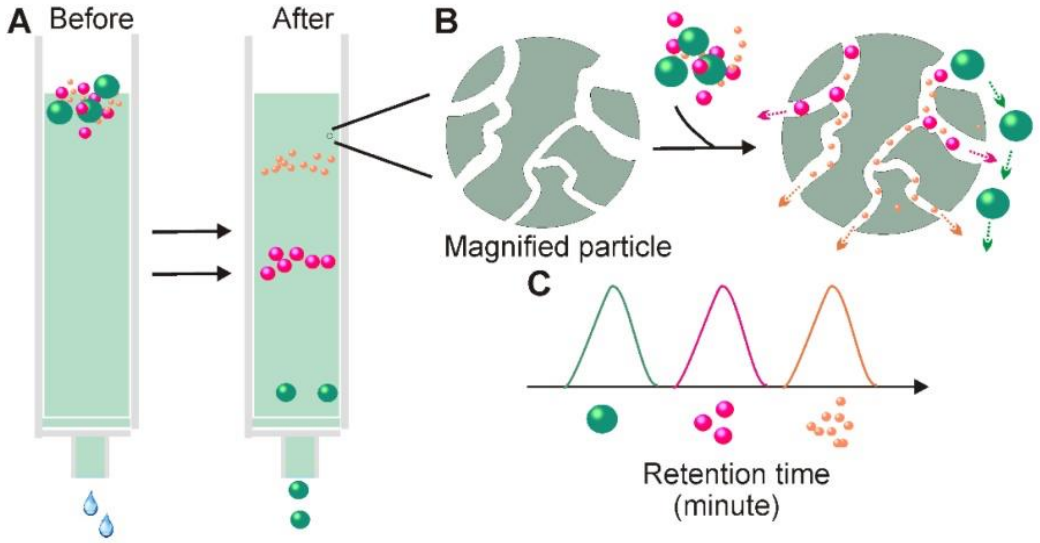
C8 and C18 chains immobilised on a silica surface



SEM image of silica beads coated with hydrocarbon carbon chains

3) Gel chromatography

- **AKA molecular-sieve chromatography**, **exclusion chromatography, gel filtration, and gel-permeation chromatography**
- Stationary phase is hydrogel beads of defined pore size, made of crosslinked **dextran**, **agarose** or **polyacrylamide**
- Migration rate depends on the size of molecules. Small molecules enter the pores of the gel beads and migrate down the column slowly. Large molecules flow past the beads, and leave the column faster



Principle of size-exclusion chromatography.
Chen et al. Theranostics 2020, Vol. 10, Issue 8

4) Ion-exchange chromatography

- Differential migration is caused by **electrostatic interactions** between the stationary phase and the target molecules

- For proteins, pH needs to be kept away from pI, so the proteins are charged

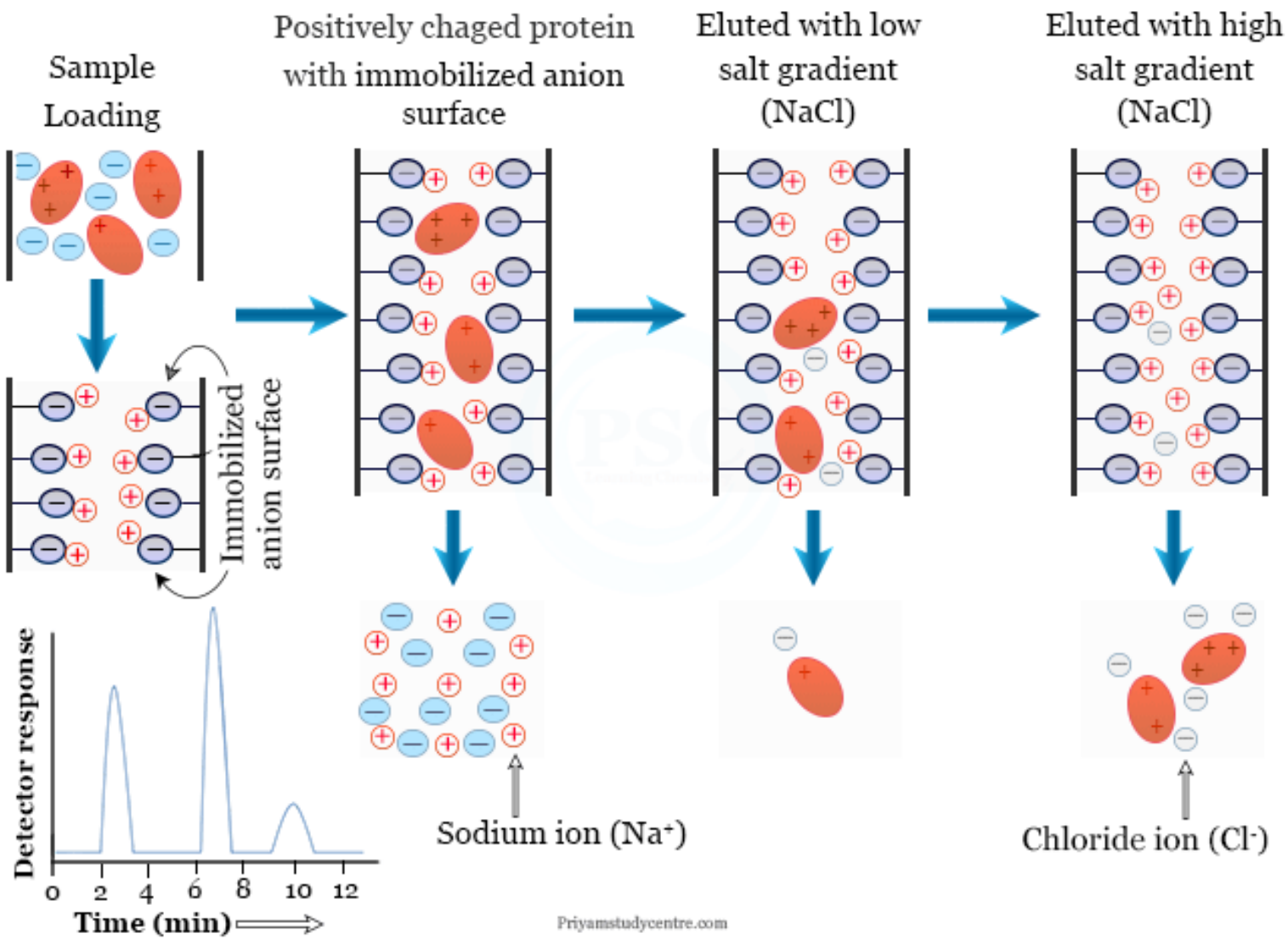
- If the stationary phase is negatively charged and the analyte is positively charged, it is called **cation-exchange chromatography**

$$S^- C^+ + A^+ B^- \rightarrow S^- A^+ + C^+ + B^-$$

Where S^- are the anionic groups on the stationary phase A^+ the target analyte, and C^+ and B^- are counterions. Here the positively charged analyte swaps places with the **exchangeable cation** C^+

- Elution can be accelerated by increasing the salt concentration (concentration of the counterions)

4) Ion-exchange chromatography



4) Ion-exchange chromatography

- If the stationary phase is positively charged and the analyte is negatively charged, it is called **anion-exchange chromatography**.

$$S^+C^- + A^-B^+ \rightarrow S^+A^- + C^- + B^+$$

Where S^+ are the cationic groups on the stationary phase A^- the target analyte, and C^- and B^+ are counterions.

- Ion exchange chromatography is used commercially for enrichment of proteins and antibiotics

5) Affinity chromatography

- Principle leading to differential migration is **specific binding between** the target compound and the stationary phase
- Uses ligands chemically connected to the stationary phase, and able to specifically bind recombinant proteins, enzymes, antibodies, hormones, nucleic acids, vitamins etc.
- **Highly specific and often used for high added value products**

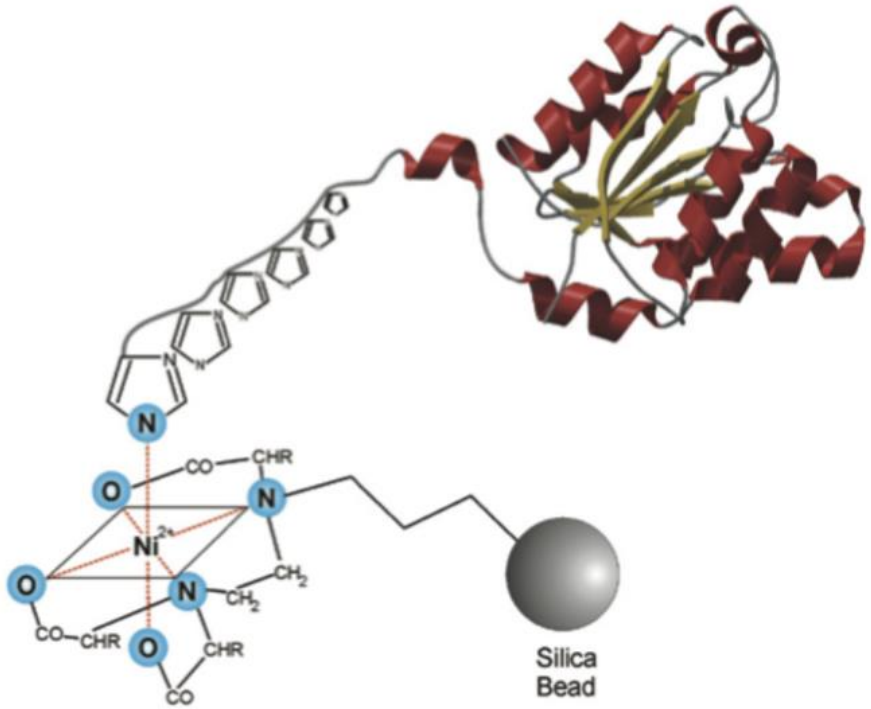
5) Affinity chromatography: Ni-NTA

- A very common affinity-chromatography approach relies on **Nitrilotriacetic acid** (NTA) as ligand.

- NTA forms a complex with **Ni^{2+}**

- The Ni-NTA complex binds strongly to **Histidine residues**

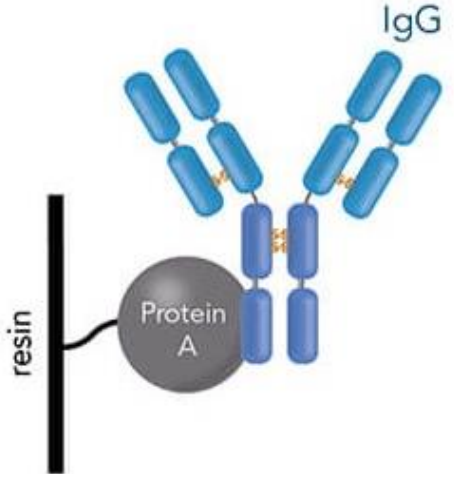
- Target recombinant proteins are purposefully modified to have a **6xHis tag** at one end, so that they can be specifically captured by the Ni-NTA complex



5) Affinity chromatography: Protein A

- Another common affinity chromatography ligand is **protein A**, used for separation of antibodies (immonoglobulins G, IgGs)

- Protein A occurs naturally on the surface of *Staphylococcus Aureus*, and can bind IgGs from a conserved region of the heavy chain that is not involved in antigen recognition. *S. Aureus* uses this trick to bind IgGs in the wrong orientation, disrupting immune response





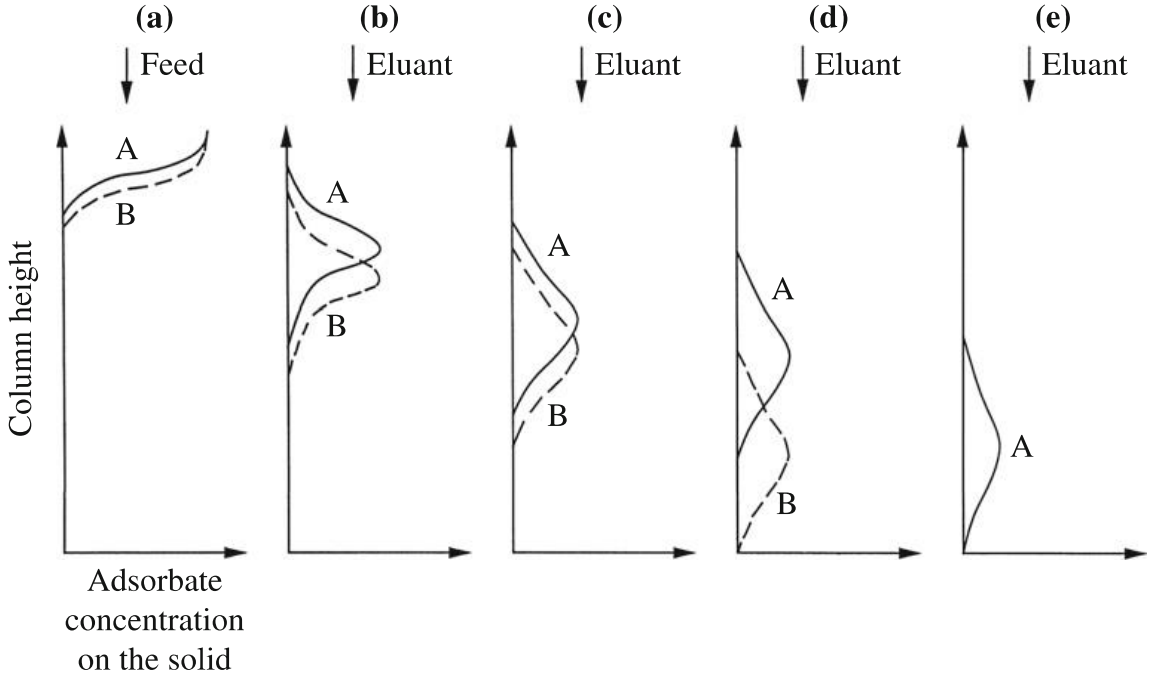
False-colour SEM image of *S. aureus.*

Differential migration

- To promote migration of solutes down the column, **eluent solutions** are passed through the bed

- The eluant promotes "binding" and "unbinding" of the solutes, at different rates

- Different **volumes of eluent**, V_e, are needed for different solutes to emerge from the column (at peak concentration), resolution in differential migration



From P.M. Doran, "Bioprocess Engineering Principles" Academic Press, 2nd ed. 2012.

Differential migration

The **capacity factor k** is also used to characterise elution

$$k = \frac{V_e - V_0}{V_0}$$

Where V_e is the elution volume and V_0 is the void volume of the column (volume not occupied by particles)

When comparing two solutes, 1 and 2, the **retention ratio** is the ratio between the capacity factors

$$\delta = \frac{k_1}{k_2}$$

Differential migration: gel chromatography

For gel chromatography

$$V_t = V_o + V_i + V_s$$

Where V_t is the total volume of the column, V_0 the void volume (between the particles), V_i the volume of the empty space within the particles, and V_s the volume of the material (hydrogel) that makes up the particles

The elution volume can be expressed as

$$V_e = V_o + K_p V_i$$

Where $\boldsymbol{K_p}$ **is the partition coefficient,** defined as the fraction of the volume within the gel particles which is available to the solute

Differential migration: gel chromatography

$$V_e = V_o + K_p V_i$$

$$K_p = \frac{V_e - V_o}{V_i}$$

- To determine K_p, we need to measure V_e and have information on V_o and V_i
- For large macromolecules that do not penetrate the gel particles, $K_p = 0$ and $V_e = V_o$, which allows us to measure V_o by measuring V_e.

Differential migration: gel chromatography

To determine V_i can use

$$V_i = a\, W_r$$

where W_r is the water retention value, defined as the volume of water taken up when rehydrating the gel per unit (dry) mass of the gel (specified by the manufacturer), and a is the dry mass of the gel.
It is often the case that the gel is provided already rehydrated, so a is unknown. In this case, we can use

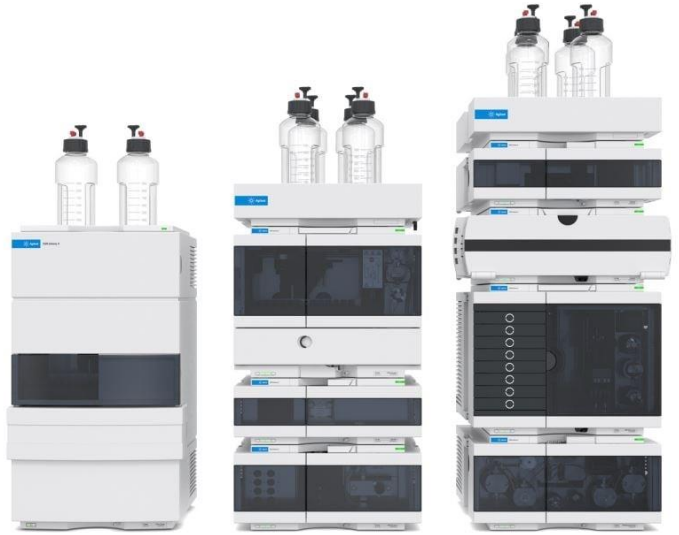
$$V_i = \frac{W_r\, \rho_g}{1 + W_r\, \rho_w} (\, V_t - V_0)$$

where ρ_g is the density of the rehydrated gel and ρ_w the density of water.

HPLC and FPLC

Two methods are used to carry out chromatographic separations of biomolecules:

1. **High Performance Liquid Chromatography (HPLC)**
   - Achieves better resolution
   - Small particles (2 to 5 μm)
   - High pressure (5-10 MPa)
   - Low throughput (1 to 5 ml min^{-1})
   - Used often at small scale or lab scale, but also for valuable products (remember oligos)



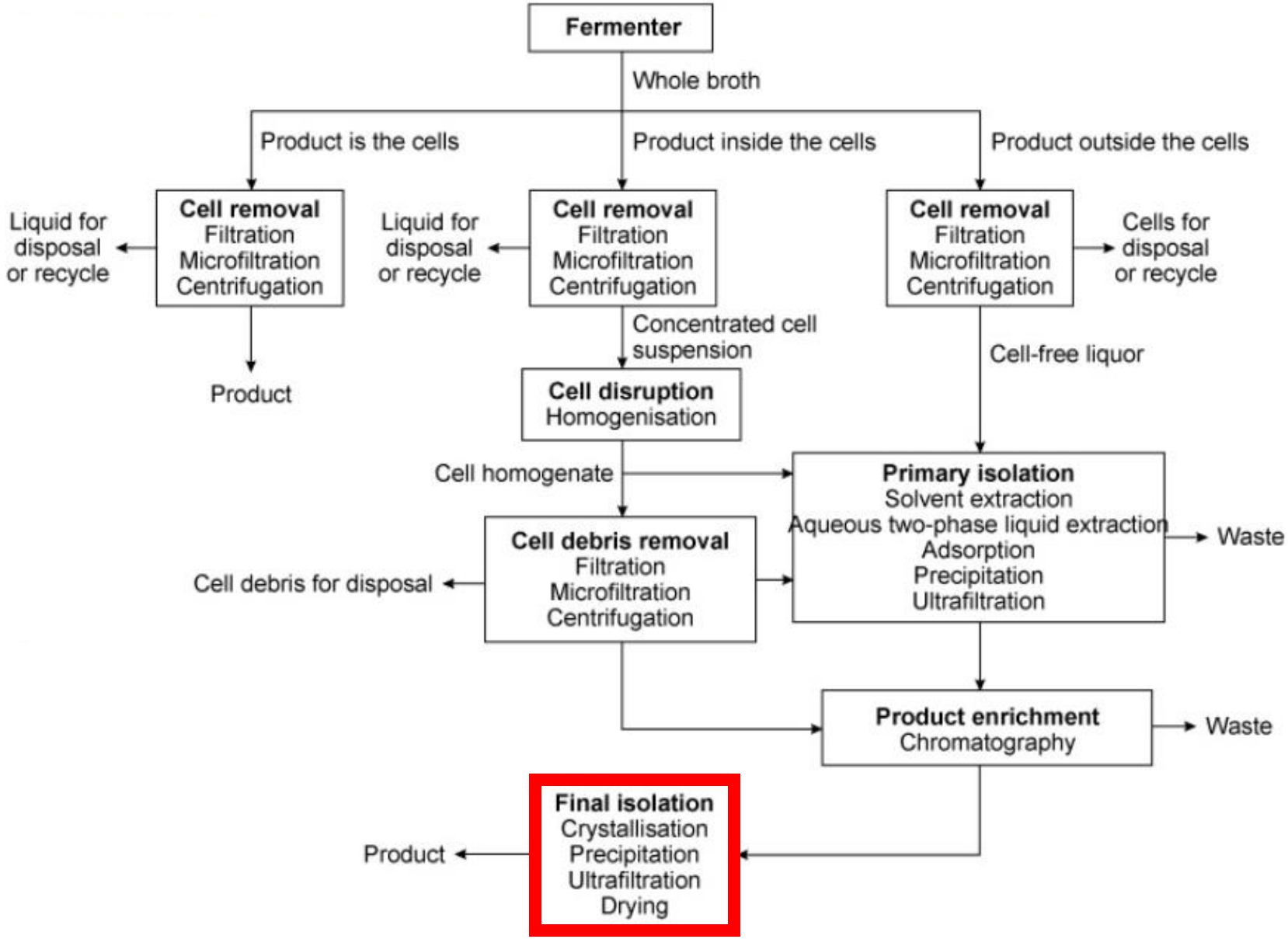
2. **Fast Protein Liquid Chromatography (FPLC)**
   - Lower resolution
   - Larger particles
   - Lower pressure
   - Higher throughput
   - Used at larger scale



Lecture outline

1) Methods for primary isolation: Adsorption
2) Product enrichment (chromatography)
   - Principle and overview of chromatography
   - Types of liquid chromatography (brief description of principles)
   - Quantify differential migration
3) **Mention of final isolation methods**

Downstream process logic



Final isolation

- Product already highly purified with chromatography
- The objective of final isolation steps is possibly **further increase purity** and alter the **physical form** of the product to make is suitable for storage and commercialization
- Many products are packaged/sold as solids/powders. These will go through either precipitation or **crystallization**, followed by filtration or centrifugation
- The wet solids (crystals, amorphous precipitate) is then **dried**
- You have seen this in previous courses!